

Designing with GaN – Miniaturization and Heating of High Frequency Converter

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First Project Design Report

Project Purpose: Development of educational demo board with the possibility of observing by students the heating of the system elements with changes in parameters such as frequency, load, voltage and gain.

Stages 1: Determining Electrical Values

- $U_{IN} = 48V$
- $U_{OUT} = \text{Adjustable}$
- $F_{sw} = \text{adjustable}$
- $L =$
 - This value has a formula [1]: $\frac{U_{OUT} \times (U_{IN} - U_{OUT})}{\Delta I_L \times F_{Sw} \times U_{IN}}$
 - $\Delta I_L = 0.3 \times I_{OUT}$ and If our $U_{OUT} = 12V$ and $P_{OUT} = 100W$ then $I_{OUT} = 8.33A$ & $\Delta I_L = 2.5A$
 - Calculation factory of “0.3” is standard value.
 - Final calculation is $L = \frac{12V \times (48V - 12V)}{2.5A \times 500 KHz \times 48V} = 7.2\mu H$

Stages 2: Component Selections

- **MCU:**
 - Texas Instruments - F28P650DK6PZPR
 - If we choose this MCU, It supports the high frequency PWM resolution. We can change the frequency and duty cycle with small movements.
 - Also this MCU has a two core. We can easily control the screen with the other core.
 - This MCU has a lots off PWM and ADC pins.
 - Price is 14.71€ per unite / 10.10€ per unite (100 Quantity)
- **GaNFET:**
 - These products are integrated power stages, meaning they have a half-bridge configuration and gate driver.
 - Texas Instruments - LMG5200
 - 80V
 - 10A
 - Max. 10 MHz Switching
 - Typical $R_{DS(ON)} = 15 \text{ mOhm}$
 - **EPC - EPC23104**
 - 100V
 - 15A
 - Max. 3 MHz Switching
 - Typical $R_{DS(ON)} = 8.7 \text{ mOhm}$

- EPC - EPC2152
 - 80V
 - 15A
 - Max. 3 MHz Switching
 - Typical $R_{DS(ON)} = 10 \text{ m}\Omega$
- Temperature Sensor:
 - Texas Instruments - TMP119AIYBGR
 - It has high sensitivity and small size.

References

- [1] B. Hauke, "Basic Calculation of a Buck Converter's Power Stage," Dec. 2011. [Online]. Available: www.ti.com